NUST Historical XLSX Extraction — Workflow

Last updated: 2026-04-28 **Validated on:** 1980 (21,261 phenotype rows, 0 flagged) **Scope:** Historical NUST XLSX reports, 1941–1986 (Green folder)

Overview

Each historical NUST XLSX file contains one year's trial report for a subset of maturity groups. The extraction pipeline sends raw cell grids to Claude (`claude-sonnet-4-6`), which interprets the table structure and returns structured JSON. No rule-based parsing is used — Claude handles all layout variation, OCR artifacts, and schema mapping.

End-to-End Pipeline

```
XLSX file(s)
    │
    ▼
scripts/extract_nust_xlsx.py        ← Step 1: extract raw tables
    │    • openpyxl reads cells
    │    • find_group_boundaries() chunks sheet by tp2 markers
    │    • each chunk sent to Claude with era-aware system prompt
    │    • JSON → flat CSVs per table type
    ▼
output_{year}/
    ├── *_phenotypes.csv
    ├── *_strains.csv
    ├── *_parentage.csv
    ├── *_descriptive.csv
    ├── *_disease.csv
    └── *_summary.csv
    │
    ▼
scripts/combine_nust_outputs.py     ← Step 2: merge multi-file years + apply
TEST_MAP
    │    • renumbers groups across files (File 2 groups offset by File 1 count)
    │    • renames Group_N → test codes (UT-I, PT-II, etc.) from TEST_MAP
    ▼
output_{year}/
    ├── combined_{year}_phenotypesTable.csv
    ├── combined_{year}_strainsTable.csv
    ├── combined_{year}_parentageTable.csv
    ├── combined_{year}_locationsTable.csv
    ├── combined_{year}_checksTable.csv
    └── combined_{year}_MetaTable.csv
    │
    ▼
fixes/fix_{year}_locs.py            ← Step 3: location table cleanup (year-
```

```
specific)
      │    • normalize state abbreviations
      │    • fix OCR city name errors
      │    • deduplicate Test×City×State
      │    • merge PlantingDate from supplemental JSON
      │    • join lat/lon from reference/nust_locations_ref.csv
      ▼
scripts/validate_nust_hist.py       ← Step 4: validate phenotypes table
      │    • schema, range, and trait-completeness checks
      │    • outputs _approved.csv and _review_flagged.csv
      ▼
output_{year}/validated/
      ├── combined_{year}_phenotypesTable_approved.csv
      └── combined_{year}_phenotypesTable_review_flagged.csv
```

After validation, the approved phenotypes table and the other combined tables are ready for the R bridge (`NUST_HistProcessing.R` → `NUST_CheckFinalFiles.R`) to produce the final `Files4Upload/` outputs.

Era-Aware System Prompt

The `SYSTEM_PROMPT` base covers the Modern era (1970–1986, validated on 1980). For earlier years, `build_system_prompt(year)` appends an era-specific addendum.

Era	Years	Addendum
Early	1941–1956	`ERA_ADDENDUM_EARLY`
Transitional	1957–1969	`ERA_ADDENDUM_TRANSITIONAL`
Modern	1970–1986	none (base prompt only)

What the addenda add

Early (1941–1956):

- Location format: City-first with dotted state abbreviation (`"Morris Minn."` → `city="Morris", state="MN"`)
- Footnote stripping from location headers (`"Guelph Ont.1"` → `"Guelph Ont."`)
- Fused tp marker handling: `tp6+7`, `tp9+10`, `tp8+11a`, `tp8+12a` — Claude splits into two phenotype sections per block
- Absent sections: tp1, tp3a, tp3b not present in early years; empty list expected
- Iodine Number of Oil trait (`IODINE NUMBER OF OIL`) — present ~1943–1948 only
- Seed Weight in centigrams (`SEED WEIGHT (cg)`) — kept separate from `SEED SIZE (g/100)`

Transitional (1957–1969):

- Mixed location formats — both modern (`"Ont. Ottawa"`) and early (`"Ottawa Ont."`) may appear in the same file
- Fused markers: `tp12a & tp12b`, `tp10 & tp12b`, `tp6+7`

- OCR normalization: `tptp11a` → tp11a, `tp 2` → tp2, `tp24` → skip, bare `tp3` → treat as tp3a if morphological
- `tp3c` (1972 only) — extract into descriptive table as-is
- Expanded multi-year summary skip list (more common in this era)

All eras (base prompt):

- Extended trait label mappings: all known variants across 46 years mapped to canonical names
- Multi-year summary skip labels: `"2-year summary"`, `"1968-70 MEAN YIELD"`, etc.
- Continuation header handling: `"(Continued)"` rows skipped, not treated as new sections

How `_normalize_tp()` works

`find_group_boundaries()` uses `_normalize_tp()` to map raw column-A cell values to canonical tp codes before boundary detection. This means the chunking logic (which splits large groups at tp6) correctly finds `tp6+7` as the first per-location marker, enabling proper sub-chunking of large early-era groups.

Raw cell value	Canonical	Notes
`tp6+7`	`tp6`	fused yield + rank
`tp9+10`	`tp9`	fused lodging + height
`tp 2`	`tp2`	spaced OCR artifact
`tp6 (Continued)`	`tp6`	continuation annotation
`tp2 (NB Baie swak beeld)`	`tp2`	Afrikaans annotation stripped
`tptp11a`	`tp11a`	doubled-prefix OCR
`tp??`	*(skip)*	unknown marker
`tp24`	*(skip)*	OCR error
`tp` (bare)	*(skip)*	separator artifact

Running the Pipeline

Single year, two XLSX files (e.g. 1980)

```
# Step 1 — extract each file
python scripts/extract_nust_xlsx.py \
  --file "input_1980/1980/Sojabone-1980 (1-89 OR).xlsx" \
  --out_dir output_1980/

python scripts/extract_nust_xlsx.py \
  --file "input_1980/1980/Sojabone-1980 (90-164 OR).xlsx" \
  --out_dir output_1980/

# Step 2 — combine and apply TEST_MAP
python scripts/combine_nust_outputs.py \
  --input_dir output_1980/ \
```

```
  --out_dir output_1980/ \
  --test_map "Group_1=UT-00,Group_2=UT-0,Group_3=UT-I,Group_4=PT-I,Group_5=UT-II,Group_6=PT-II,Group_7=UT-III,Group_8=PT-III,Group_9=UT-IV,Group_10=PT-IV"

# Step 3 — location cleanup (year-specific fix script)
python fixes/fix_1980_locs.py

# Step 4 — validate
python scripts/validate_nust_hist.py \
  --input output_1980/combined_1980_phenotypesTable.csv \
  --out_dir output_1980/validated/
```

Batch run (all XLSX files in a folder)

```
python scripts/extract_nust_xlsx.py \
  --dir "R:/NUST_Historical_Data/Green-.../Green/1952/" \
  --out_dir output_1952/
```

The era is detected automatically from the year in the filename. The active era and prompt length are printed at the start of each file:

```
Processing: Sojabone-1952 (1-60 OR).xlsx
  Year detected: 1952
  Era: early (system prompt: 9417 chars)
  Sections found: ['GlobalParentage', 'Group_1', 'Group_2', ...]
```

Identifying the TEST_MAP for a year

If the Group→TestCode mapping is not known, use `extract_test_map_pdf.py` with the source PDF for that year:

```
python scripts/extract_test_map_pdf.py \
  --pdf "input_{year}/{year}_source.pdf" \
  --out_dir output_{year}/
```

Outputs a JSON file and an R snippet ready to paste into `combine_nust_outputs.py`.

Output Tables

File	Content	Format
`combined_{year}_phenotypesTable.csv`	Strain × Location × Trait values	long (one row per observation)

File	Content	Format
`combined_{year}_strainsTable.csv`	Strain, generation, previous testing	wide
`combined_{year}_parentageTable.csv`	Strain, Female × Male cross	wide
`combined_{year}_locationsTable.csv`	City, State, lat, lon, PlantingDate, Conductor	wide
`combined_{year}_checksTable.csv`	Check entries, RM (NULL for pre-~1985)	wide
`combined_{year}_MetaTable.csv`	Per-location metadata (CV, LSD, reps)	wide
`combined_{year}_maturityVerification.csv`	Maturity date → DOY conversion	wide

Validation Rules

`scripts/validate_nust_hist.py` checks:

- **Schema:** all required columns present
- **Range:** numeric traits within plausible bounds
- **Non-numeric:** unexpected string values in numeric phenotype columns
- **Trait completeness:** expected traits present in every test group (with known omissions per era)
- **Strain count consistency:** uniform strain count across traits within each group (tolerance ±5)

Known trait omissions (suppressed from completeness warnings):

Test pattern	Exempt traits	Reason
`PT-*`	YieldRank	Preliminary tests never assigned ranks
`PT-III`	YieldRank, Maturity	MG III prelim omits maturity
`UT-0`	YieldRank	Not reported (1980 confirmed)
`UT-III`	YieldRank	Not reported (1980 confirmed)

> Additional exemptions may need to be added for other years — confirm against source PDF before adding to `KNOWN_TRAIT_OMISSIONS`.

Reference Files

File	Purpose
`reference/nust_locations_ref.csv`	115 geocoded locations (1941–1986); NeedsVerification=22

File	Purpose
`reference/2024_NUST_Locations_PlotInfo.csv`	Modern trial GPS (43 locations; identical to 2025 file)
`docs/system_prompt_multiyr_notes.md`	Detailed notes behind era addenda; open questions
`logs/NUST_Batch_Cost_Estimate.md`	API cost projections by era
`logs/NUST_1980_Open_Items.md`	Documented data gaps for 1980 (reference for other years)

Known Gaps and Remaining Work

Item	Notes
Era-aware prompt not yet tested on pre-1970 XLSX	Run a test year from each era (~$30 total) before full batch
1975	PDF-only — no XLSX in Green folder; separate extraction path needed
1987–1988	Fragmented XLSX structure; separate assembler script needed
Historical MG → RM table	Needed to populate checksTable RM for pre-~1985 years; RM NULL until resolved
Conductor field	Not in 1980 publication; verify availability in other years
Seed Weight (cg) unit	Pre-~1968 seed size in cg, not g/100 — confirm crossover year; may need unit conversion for unified analysis
Iodine Number	Last year to verify (~1948); not relevant after that
tp3c (1972)	Inspect that file to determine what data is present
Open questions	See `docs/system_prompt_multiyr_notes.md` Section 7